Goal: Prove that RSA actually works.

More formally:

Thm Consider an RSA cryptosystem with public key (n,e) and corresponding private key (n,d).

Then

$$\boxed{m^{de} \equiv m \pmod{n}}$$

for all $m \in \mathbb{Z}$.

The proof requires some preparation.

Reminder: for $n \geq 1$, Euler's totient function φ is defined via

$$\varphi(n) := \text{no. of those } i = 1, \dots, n \text{ with } \gcd(i,n) = 1$$
$$= |\mathbb{Z}_n^\times| \quad \text{(see lecture 5)}$$

Ex:

- $\varphi(4) = 2$ 1 2 3 4
- $\varphi(5) = 4$ 1 2 3 4 5
- $\varphi(12) = 4$ 1 2 3 4 5 6 7 8 9 10 11 12

Chinese remainder theorem:

Let $m, n \geq 1$ with $\gcd(m,n) = 1$ and let $a, b \in \mathbb{Z}$.
Then there exists an $x \in \mathbb{Z}$ with $x \equiv a \pmod m$
and $x \equiv b \pmod n$.

Any two numbers with this property are congruent modulo mn.

For a constructive, classical proof, see e.g. Hardy & Wright, Ch VIII

Sketch of a non-constructive proof:

Define a function

$$f: \mathbb{Z}_{mn} \longrightarrow \mathbb{Z}_m \times \mathbb{Z}_n,$$

$$= \{(y,z) : y \in \mathbb{Z}_m, z \in \mathbb{Z}_n\}$$

$$x \longmapsto (x \bmod m, \ x \bmod n).$$

Then f is injective. Indeed, suppose that $f(x) = f(x')$ for $x, x' \in \mathbb{Z}_{mn}$. Then $x \equiv x' \pmod m$ and also $x \equiv x' \pmod n$. That is, m and n both divide

$x - x'$.

Since $\gcd(m,n) = 1$, the product mn then divides $x - x'$. That is, $x \equiv x' \pmod{mn}$ and therefore $x = x'$.

This proves that f is injective. Since $|\mathbb{Z}_{mn}| = mn = |\mathbb{Z}_m \times \mathbb{Z}_n|$ is finite, f is thus bijective. The claim follows easily. //

The Chinese remainder theorem has the following important consequence:

Thm For $\gcd(m,n)=1$, we have $\boxed{\varphi(mn)=\varphi(m)\varphi(n).}$

Sketch of proof:

First, a definition: a reduced residue system (RSS) modulo n is a set $R\subset\mathbb{Z}$ with the following properties:

- $\gcd(r,n)=1$ $\forall r\in R$
- For each $a\in\mathbb{Z}$ with $\gcd(a,n)=1$, there exists a unique $r\in R$ with $a\equiv r \pmod n$.

Ex: $\mathbb{Z}_n^\times=\{a\in\mathbb{Z}_n:\gcd(a,n)=1\}$ is a RRS mod n.

Easy: given RRSs R and R' mod n, there exists a (unique) bijection $f:R\to R'$ such that

$$f(r)\equiv r \pmod n$$

for each $r\in R$.

In particular: $\varphi(n) = |R|$, where R is any RRS mod n.

Let $R(m)$ and $R(n)$ be RRSs mod m and n, resp.

By the Chinese remainder theorem, for $r \in R(m)$ and $s \in R(n)$, there exists a (unique) $x_{rs} \in \mathbb{Z}_{mn}$ with

$$x_{rs} \equiv r \pmod{m}$$

and

$$x_{rs} \equiv s \pmod{n}$$

Easy/exercise: $X := \{x_{rs} : r \in R(m), s \in R(n)\}$ is a RRS mod mn consisting of $|R(m)| \cdot |R(n)|$ elements.

Hence:
$$\varphi(mn) = |X| = |R(m)| \cdot |R(n)|$$
$$= \varphi(m)\,\varphi(n).$$
$_{/\!/}$

(For algebraists:)

Second proof:

The Chinese remainder theorem yields a ring isomorphism

$$\mathbb{Z}_{mn} \xrightarrow{\cong} \mathbb{Z}_m \times \mathbb{Z}_n .$$

It induces an isomorphism of unit groups

$$\mathbb{Z}_{mn}^{\times} \xrightarrow{\cong} (\mathbb{Z}_m \times \mathbb{Z}_n)^{\times} \overset{!}{=} \mathbb{Z}_m^{\times} \times \mathbb{Z}_n^{\times} .$$

Hence, $\varphi(mn) = |\mathbb{Z}_{mn}^{\times}| = |\mathbb{Z}_m^{\times}| \cdot |\mathbb{Z}_n^{\times}| = \varphi(m)\varphi(n).$

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Exercise: Let p be a prime.

Then $\boxed{\varphi(p^a) = p^a - p^{a-1} = (p-1)p^{a-1}}$

for $a \geqslant 1$. In particular, $\varphi(p) = p-1$.

Ex:

$$\varphi(45) = \varphi(9 \cdot 5) \underset{\gcd(9,5)=1}{=} \varphi(9) \cdot \varphi(5)$$

$$= \varphi(3^2) \cdot \varphi(5^1)$$

$$\underset{3,5 \text{ prime}}{=} (3^2 - 3) \cdot (5 - 1)$$

$$= 6 \cdot 4$$

$$= 24$$

We'll use the following theorem to prove that RSA works.

Euler's Totient Theorem (Euler 1763)

Let $n \geq 1$ and let $a \in \mathbb{Z}$ with $\gcd(a,n) = 1$.

Then:

$$\boxed{a^{\varphi(n)} \equiv 1 \pmod{n}.}$$

Ex: $\gcd(2,45)=1$ and $\varphi(45)=24$
(see above)

$$2^{24} = 16{,}777{,}216 \overset{!}{=} 45\cdot 372{,}827+1$$

so $$2^{24}\equiv 1 \pmod{45},$$

as predicted by Euler's Totient Theorem.

Sketch of proof of the theorem:

Let $R=\{r_1,\dots,r_{\varphi(n)}\}$ be a RRS mod n.

Exercise: since $\gcd(a,n)=1$,
$$R' := \{ar_1,\dots,ar_{\varphi(n)}\}$$

is also a RRS mod n.

Hence: $\exists$ bijection $f:\{1,\dots,\varphi(n)\}\to\{1,\dots,\varphi(n)\}$

such that

$$\boxed{a r_i \equiv r_{f(i)} \pmod{n}}$$

for all $i = 1, \dots, \varphi(n)$.

Therefore:

$$a^{\varphi(n)} \cdot \prod_{i=1}^{\varphi(n)} r_i = \prod_{i=1}^{\varphi(n)} a r_i \equiv \prod_{i=1}^{\varphi(n)} r_{f(i)}$$

$$\equiv \prod_{i=1}^{\varphi(n)} r_i \pmod{n}.$$

Easy: $\gcd(r_1 \cdots r_{\varphi(n)}, n) = 1$

Upon multiplying by a modular inverse of $\prod_{i=1}^{\varphi(n)} r_i$, the congruence

$$a^{\varphi(n)} \cdot \prod_{i=1}^{\varphi(n)} r_i \equiv \prod_{i=1}^{\varphi(n)} r_i \pmod{n}$$

thus implies

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

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(For algebraists:)

Second proof:

If G is any finite group and $g \in G$, then the order of g divides $|G|$ by Lagrange's theorem.

Hence:

$$g^{|G|} = 1.$$

Taking $G = \mathbb{Z}_n^\times$, for $a \in \mathbb{Z}_n^\times$, we obtain

$$a^{\varphi(n)} = 1$$

in $\mathbb{Z}_n^\times$. //

We'll use the following consequence of Euler's Totient Theorem:

Fermat's little theorem: (Stated: Fermat 1640. Proved: Euler 1736)

Let p be a prime and let a be an integer which is not divisible by p. Then:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}.}$$

Pf: • p prime $\Rightarrow$ $\varphi(p) = p-1$

• a is not divisible by $p \Longleftrightarrow \gcd(a,p) = 1$

• Euler's Totient Thm $\Longrightarrow$ $a^{p-1} = a^{\varphi(p)} \equiv 1 \pmod{p}$.

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Remark/exercise: If p is a prime, then

$$a^p \equiv a \pmod{p}$$

for all $a \in \mathbb{Z}$. This easily follows from Fermat's little theorem.

We can now, finally, show that RSA works:

Thm Consider an RSA cryptosystem with public key (n,e) and corresponding private key (n,d). Let $m \in \mathbb{Z}$ be arbitrary. Then: $\boxed{m^{de} \equiv m \pmod{n}}$

Hence: if $m \in \mathbb{Z}_n$ and $c = m^e \bmod n$, then $m = c^d \bmod n$.

Pf:

Recall that $n = pq$, where p and q are two different (!) primes.

Since $p \neq q$, the one congruence

$$m^{de} \equiv m \pmod{n}$$

is equivalent to the two simultaneous congruences

$$m^{de} \equiv m \pmod{p}$$

and $$m^{de} \equiv m \pmod{q}.$$

Consider the first of these. Two cases:

① $p \mid m$: Then: $m^{de} \equiv 0 \equiv m \pmod{p}$.

② $p \nmid m$: Since $de \equiv 1 \pmod{(p-1)(q-1)}$, there exists an integer $x > 0$ with

$$\underbrace{de}_{\geq 1} = 1 + (p-1)(q-1)x.$$

By Fermat's little theorem,

$$m^{p-1} \equiv 1 \pmod{p}$$

and thus

$$m^{(p-1)(q-1)x} \equiv \left(m^{p-1}\right)^{(q-1)x} \equiv 1 \pmod{p}.$$

Hence:

$$m^{de} = m^{1+(p-1)(q-1)x}$$

$$= m \cdot m^{(p-1)(q-1)x}$$

$$\equiv m \cdot 1 \equiv m \pmod{p}.$$

In the same way:

$$m^{de} \equiv m \pmod{q}.$$

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